29

Empyema

Introduction

Parapneumonic effusion (PPE) can occur with pneumonia. All PPE do not need specific treatment. According to the characteristics, intervention is decided.

Parapneumonic effusion: Collection of fluid in pleural space in association with underlying lung infection.

Uncomplicated parapneumonic effusion (UPPE): Effusion is sterile—low white cell count and free flowing fluid without any septations.

Complicated parapneumonic effusion (CPPE): Invasion of bacteria into pleural space—significant increase in pleural fluid white cell count and deposition of fibrin. Septations and loculations present.

Empyema: Late stage of CPPE—grossly purulent fluid in pleural cavity ± presence of bacterial organisms (Gram stain or culture).

Etiology—Common Organisms

Immunocompetent: *Streptococcus pneumoniae*, *Staphylococcus aureus* [methicillin-susceptible *S. aureus* (MSSA) and methicillin-resistant *S. aureus* (MRSA)], *Streptococcus pyogenes*, *Haemophilus influenzae* (rare after UIP)].

Immunocompromised: Gram-negative organisms (*Pseudomonas aeruginosa* and *Klebsiella*), fungi depending on the local hospital's prevalence of type of organism.

Pathogenesis

There are three stages and all three stages are a continuum:

1. **Exudative phase (first 1–3 days of PPE):** Fluid is thin with minimal cellular response with pH > 7.3, glucose > 60 mg/dL and lactate dehydrogenase (LDH) < 1,000 IU/L. Gram stain and culture are negative for microorganism.
2. **Fibrinopurulent phase (day 4–14 of PPE):** Pleural fluid shows large number of polymorphonuclear leukocytes and fibrin. With continued accumulation of neutrophils and fibrin, PPE becomes purulent and viscous leading to development of empyema. Pleural fluid is purulent with pH < 7.2, glucose < 40 mg/dL, and LDH >1,000 IU/L. Gram stain and culture shows microorganism.
3. **Organization phase (after 14 days):** Fibroblasts grow into exudates on both visceral and parietal pleural surfaces, producing an inelastic membrane "the peel" that entraps lung and prevent expansion.

Presentation

- ☑ Suspect empyema if child admitted with pneumonia fails to respond 48 hours after initiation of appropriate antibiotic therapy.
- ☑ Child appears ill with high-grade fever, malaise, loss of appetite, breathlessness, cough, and chest pain. May lie on affected side to splint hemithorax for temporary analgesia.
- ☑ On examination, sick look, respiratory distress, reduced chest movement and expansion, dull on percussion, reduced or absent breath sounds, and scoliosis on the affected side.

Complications

If left untreated may result in atelectatic lung, bronchopleural fistula, and rarely empyema necessitans.

Clinical features/imaging [chest X-ray (CXR), ultrasonography (USG) chest, and contrast-enhanced computed tomography (CECT) chest]/blood and pleural fluid investigations.

Blood Investigations

Complete blood count (↑ TC), C-reactive protein (CRP) trend, electrolytes [for syndrome of inappropriate antidiuretic hormone secretion (SIADH)], blood culture (low positivity), total proteins, and albumin

Pleural Fluid Investigations

Cell count [↑↑ white blood cell (WBC)], pH (<7.2), LDH (>1,000 U/L), proteins, glucose (<40 mg/dL), Gram stain, culture, and polymerase chain reaction (PCR)

Chest X-ray

Uniform opacity of hemithorax with mediastinal shift to opposite side, serial CXR—useful

Ultrasonography

Identify nature of fluid (loculated/free flowing), septations, volume, and shows optimal site for thoracentesis. Serial ultrasound (US) helpful.

Contrast CT Chest

Identifies underlying lung parenchymal necrosis/abscess, thickness of pleura, and helps surgeon plan for surgical treatment. Identifies collection in mediastinal aspect of pleural cavity, associated pericardial effusion, and mediastinal adenopathy but does not identify septations.

Management

TABLE 1: Empyema management (imaging-based stage of disease).

Stage of disease	***Treatment***
Stage I (exudative phase): Effusion—low cellularity, bacteria free, clear free flowing fluid—US	Antibiotics, monitoring for worsening ± ICD if no clinical improvement but increasing pleural fluid
Stage II (fibrinopurulent phase): Pleural fluid—high cellularity, bacterial presence, echogenic fluid—US—septation/loculation	Antibiotics + ICD + fibrinolytics/VATS
Stage III (organizational phase): Reorganization of fluid—solid fibrous peel/thick membrane—prevent lung expansion	VATS/Mini-thoracotomy

(ICD: intercostal drainage; US: ultrasound; VATS: video-assisted thoracoscopic surgery)

Antibiotics

- ☑ Cover common organisms (*Streptococcus pneumoniae* and *Streptococcus aureus*): Antibiotics: Ceftriaxone + cloxacillin.
- ☑ Modify as per culture results. Initially intravenous (IV) route. Duration of antibiotics: 2–4 weeks—change from IV to oral—based on improvement in general well-being with decrease in trend of inflammatory markers. Some may need vancomycin/linezolid if methicillin-resistant *Staphylococcus aureus* (MRSA) on culture or if hemodynamically unstable. Addition of clindamycin may be required specifically if associated soft tissue involvement.

Supportive Management

- ☑ Paracetamol for fever, analgesics for pain due to intercostal drainage (ICD), IV fluids, and oxygen supplementation—nasal prongs/high-flow nasal cannula/noninvasive ventilation (NIV) based on respiratory distress severity.
- ☑ Exercises that help in lung expansion (incentive spirometry and deep breathing exercise) may improve drainage through ICD. There is no role for chest physiotherapy or inhaled bronchodilators.

Intercostal Drainage (for pH < 7.2)

- ☑ With under water seal drainage. Small bore ICD gives same result as large bore ICD. If available, pig tail catheter can be used.
- ☑ Ultrasound-guided ICD placement ideal. Midaxillary line 5/6th intercostal (IC) space on affected side. Negative suction not required. Daily monitoring of ICD fluid needed. ICD removal if fluid drained < 1–2 mL/kg/day (or <10–20 mL/day) for 2–3 consecutive days + clinical and CXR/US improvement.

Fibrinolytics

- ☑ *Indication*: Loculations/septations in US—early in course of empyema till 7–10 days. Recent studies show fibrinolysis effective with more doses and even beyond 14 days.
- ☑ Instilled through ICD and mobilization encouraged to aid dispersion.
- ☑ *Adverse effects*: Fever, intrapleural bleeding, anaphylaxis, discomfort during intrapleural injection, and transient blood staining of drainage fluid.
- ☑ *Absolute contraindication*: Bronchopleural fistula and bleeding diathesis.
- ☑ *Predictors of failure*: Pleural thickening, intraparenchymal abscess, and necrotizing pneumonia.

Urokinase	Streptokinase
☑ <1 year: 10,000 U in 10 mL NS	☑ <1 year: 10,000 units/kg
☑ >1 year: 40,000 U in 40 mL NS	☑ >1 year: 20,000 units/kg
☑ 4 hours dwell time—instillation	☑ Dissolved in 50 mL NS
☑ Twice daily—12 hours apart	☑ 4 hour dwell time—infusion over 1 hour
☑ Six instillations in 3 days	☑ Three consecutive days
	☑ Four to six doses—superior to three doses

Management

Surgical Intervention

- ☑ When sepsis and infected fluid not effectively controlled with antibiotics and ICD and/or presence of significant respiratory compromise due to thickened pleura.
- ☑ If ICD drainage output decreases but clinical deterioration present with persistence of effusion by imaging.

Surgical Procedures

Either of the following may be chosen (depending on availability):

- ☑ Video-assisted thoracoscopic surgery (VATS) (rare complication—trauma to lung parenchyma and persisting postoperative air leak)
- ☑ Mini-thoracotomy/decortication

Fibrinolytics or Video-assisted Thoracoscopic Surgery

- ☑ Equipoise between VATS and fibrinolytics
- ☑ Centers with minimally invasive surgical expertise: VATS—better option. Shorter postoperative hospital stay. Reduced need for reintervention.
- ☑ Centers lacking minimally invasive surgical expertise: Fibrinolytics

Respiratory Outcomes

Good long-term outcome (clinical as well as by pulmonary function test) irrespective of type of intervention (medical or surgical) and age at illness.

Further Reading

- ☑ de Benedictis FM, Carloni I, Osimani P, Cobellis G, Martino A, Lanza C, et al. Prospective evaluation of lung function in children with parapneumonic empyema. Pediatr Pulmonol. 2019;54:421-7.
- ☑ Derderian SC, Meier M, Partrick DA, Demasellis G, Reiter PD, Annam A, et al. Pediatric empyemas—Has the pendulum swung too far? J Pediatr Surg. 2020;55:2356-61.
- ☑ Mathew JL, Soni V, Singh M, Mittal P, Singhi S, Gautam V, et al. Intrapleural streptokinase is effective and safe for children with multi-loculated empyema regardless of the time from disease onset. Acta Paediatr. 2018;107:2165-71.
- ☑ Pacilli M, Nataraja RM. Management of paediatric empyema by video-assisted thoracoscopic surgery (VATS) versus chest drain with fibrinolysis: systematic review and meta-analysis. Paediatr Respir Rev. 2019;30:42-8.
- ☑ Shankar G, Sahadev R, Santhanakrishnan R. Pediatric empyema thoracis management: should the consensus be different for the developing countries? J Pediatr Surg. 2020;55:513-7.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Persistent Diarrhea

Lead Author
Riyaz A

Co-Authors
Kosha Gajera, Vishnu D Biradar

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

© Indian Academy of Pediatrics

IAP Standard Treatment Guidelines Committee

Chairperson

Remesh Kumar R

IAP Coordinator

Vineet Saxena

National Coordinators

SS Kamath, Vinod H Ratageri

Member Secretaries

Krishna Mohan R, Vishnu Mohan PT

Members

Santanu Deb, Surender Singh Bisht, Prashant Kariya, Narmada Ashok, Pawan Kalyan